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Simple and low-cost tabletop NMR system for chemical-shift-resolution spectra measurements

Y. Hibino[a,*], K. Sugahara[a,b], Y. Muro[c], H. Tanaka[c], T. Sato[c], Y. Kondo[a,d]

[a]Graduate School of Science and Engineering Research, Kindai University, 577-8502 Higashi-Osaka, Japan
[b]Department of Electronics and Engineering, Kindai University, 577-8502 Higashi-Osaka, Japan
[c]THAMWAY Co., Ltd, 417-0001 Fuji, Japan
[d]Department of Physics, Kindai University, 577-8502 Higashi-Osaka, Japan

Abstract

We have been working on developing a low-cost tabletop NMR system. We reported that a field homogeneity as high as 50 ppm was achieved with a simple NMR magnet by employing two facing ferrite magnets with iron disks in between (JMR **275** (2017) 114-119) . In this paper, we report two improvements added to our previous system: (1) an FPGA based signal processing unit to improve the S/N ratio and (2) a simple shimming mechanism to improve the field homogeneity. We obtained as high as 1 ppm field homogeneity in the best case. The signals from hydrogen nuclear spins in a methyl and carboxy group in acetic acid were resolved in NMR spectra.

Keywords: NMR, Education, Ferrite Magnets, FPGA, Shimming

1. Introduction

High resolution nuclear magnetic resonance (NMR) [1] is widely employed in the field of drug discovery. Superconducting NMR magnets are generally employed to generate a strong and homogeneous magnetic field in the sample space since both high sensitivity and good frequency resolution are required in order to investigate large molecules such as enzymes. However, these NMR systems are large and expensive, and are available only at specific facilities. In addition to superconducting NMR systems, various research groups have been developing compact NMR systems using permanent magnets and have applied them for various works [2–4].Halbach arrays of permanent magnets [5] have often been employed for these compact NMR systems [6–13] in order to obtain strong magnetic fields.

We, however, have been focusing on a simpler NMR magnet with two ferrite magnets facing each other among the above compact NMR systems for academic teaching in mind [14, 15]. Our system is not simple but cost-effective like other reported low cost NMR systems or magnets [7, 8, 13, 16–21]. The magnetic field generated by two facing ferrite magnets is not ideally magnetized although it is largely improved by adding iron disks in between [22]. We have *accidentally* succeeded to compensate these differences a little bit by arranging another small magnets outside of the ferrite magnets. In this paper, we show our attempt of placing magnetic pieces as reproducible as possible: we put iron screws on a plate with threaded holes. Note that small magnetic pieces were successfully introduced in order to improve the field homogeneity of NMR magnets [11, 23–25]. We also introduce an FPGA based signal processing unit such as in [20, 26–29] in order to improve the S/N ratio of our system. This improvement is necessary in order to overcome the poor temperature stability of ferrite magnets.

With these improvements, the homogeneity of the magnetic field can be further improved from 50 ppm to 1 ppm in the best case.

2. NMR system

2.1. FPGA based signal processing unit

The followings are our previous measurement electronics [22]. The signal from the pickup coil was amplified by a low noise preamplifier (CA-251F4: NF Circuit Design) and directly measured with an oscilloscope (MOD3012: Tektronix) without converting the frequency. We employed this electronics for simplicity at the cost of the bad S/N ratio. Note that the preamplifier is wide-band (0 ∼ 10 MHz). It was essential to average the signal in order to observe an FID signal.

We introduced an FPGA (Cyclone III, "EP3C16E144C8N" running at 80 MHz clock) based signal processing unit in order to improve the S/N ratio of our system. The unit is a compact single board with dimensions of 125 mm × 75 mm, as shown in Fig. 1. This board generates an rf pulse sequence, such as a CPMG sequence [1], and provides the function of a lock-in amplifier [30]. See the supplemental material for more details of the FPGA board: we provided more detailed circuit diagrams and programs for configuring the chip. Although we employed the same preamplifier, this lock-in amplifier function reduced the effective bandwidth of the preamplifier: it can be as narrow as 1 kHz in stead of 10 MHz. Therefore, this board allows us

*Corresponding author
Email addresses: 1733340441n@kindai.ac.jp (Y. Hibino), ksugahar@kindai.ac.jp (K. Sugahara), ykondo@kindai.ac.jp (Y. Kondo)

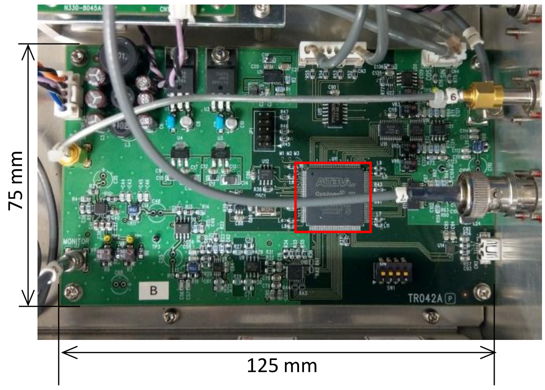

Figure 1: FPGA based signal processing unit. The LSI marked with the red square is a FPGA chip.

to measure an FID signal without averaging. The capability of a single FID detection was essential in this work. We needed a sample of which T_2 is of the order of 1 s, and thus, its T_1 must also be the same order, in order to confirm the 1 ppm field homogeneity with ferrite magnets. They have a temperature dependence that is as bad as $-0.18\%/^\circ$C. The waiting time of a few seconds, which was required by the long $T_1 \sim 1$ s of this sample, should cause an excessively large field change during an FID signal averaging: we could not average signals. Note that the temperature of the NMR magnet was not controlled, since we were afraid that naive temperature control might cause serious hunting.

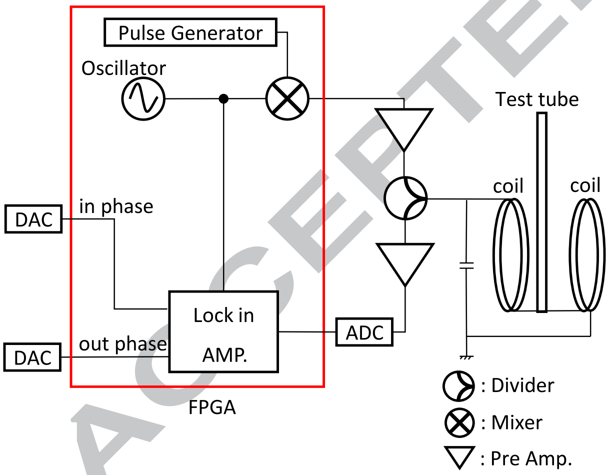

Figure 2: Block diagram of the FPGA based signal processing unit.

2.2. NMR magnet

The left panel of Fig. 3 is an overview of our reported tabletop-NMR magnet [22] and the right panel is a sketch of its structure. See [22] for more details. The field homogeneity of this NMR magnet was improved by attaching several iron screws as a shimming mechanism (see Fig. 4). We designed a shim holder in order to make the shimming procedure as reproducible as possible. The details of the shimming procedure are discussed in the next section. The shim holder was made of PLA resin by

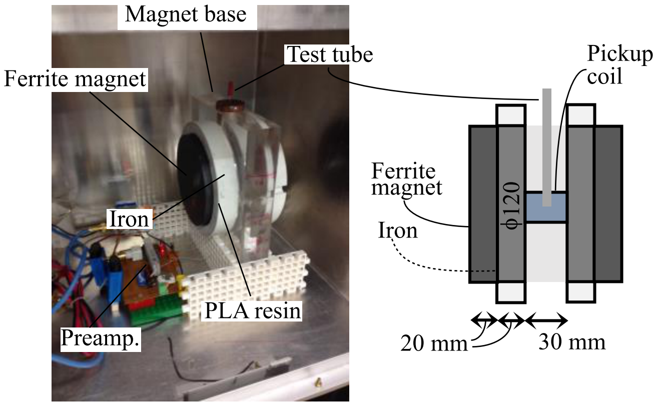

Figure 3: Sketch of a reported NMR magnet [22]. The left panel shows the NMR magnet, termed as “homemade NMR magnet”, and the electronics for measurement. The right panel shows its structure.

using a 3D printer. It has 61 screw holes in a triangular lattice whose lattice constant is 15 mm.

The magnetic field changed significantly when the room temperature varied because of the large temperature dependence of ferrite magnets. Therefore, we adjusted the frequency origin in the spectra in this study so that the water peak was at 4.7 ppm, which is a reference value for pure water. Note also that the rf pulse frequency, which is equal to the reference frequency of the lock-in amplifier in the FPGA data processing unit, was also adjusted according to the room temperature. The samples were put in NMR test tubes having a diameter of 3 mm.

3. Shimming NMR magnets with iron screws

3.1. Shimming procedure

If ferrite magnets were perfect without any defects unlike our magnets [22], the generated magnetic field ought to have been ideally homogeneous. If we could know our magnets perfectly, including the magnetization distribution inside the magnets, we might be possible to calculate how to compensate it. Although we have been working in this direction with a truncated SVD method [31], we have not yet reached this level of

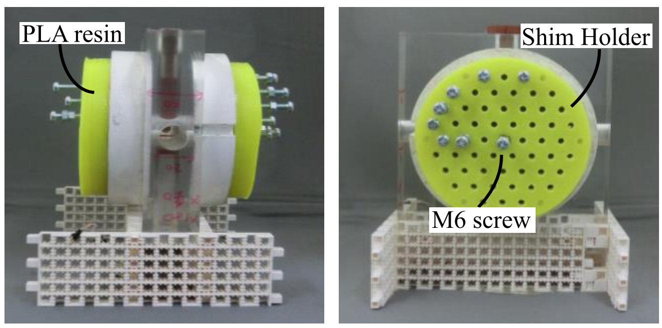

Figure 4: Shim holders placed on the homemade NMR magnet shown in Fig. 3.

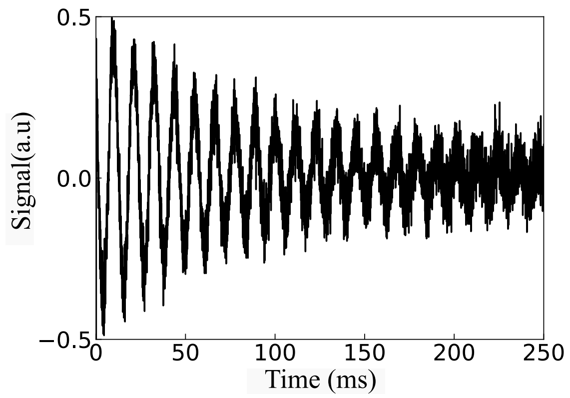

Figure 5: FID signal with homemade NMR magnet shown in Fig. 4. No averaging was required. Sample was pure water in a 3 mm diameter NMR test tube. The rf pulse frequency (= reference frequency of lock-in amplifier in the FPGA data processing unit) = 2.5041 MHz (58.813 mT). The cutoff frequency of the lock-in amplifier = 10 kHz. We converted the frequency of the lock-in amplifier to the field strength with 1 T ⇔ 42.577478 MHz.

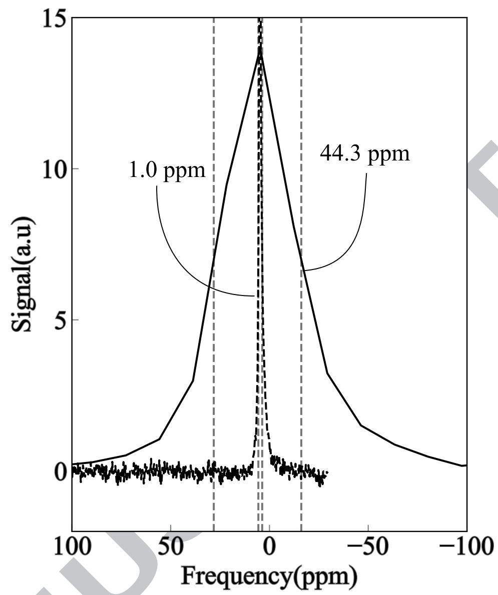

Figure 6: Spectra of pure water in a 3 mm NMR test tube measured with homemade NMR magnet. Solid line is before shimming (FWHM: 44.3 ppm), and dotted line is after shimming (FWHM: 1.0 ppm). The rf pulse frequency = 2.4500 MHz (57.542 mT). The cutoff frequency of the lock-in amplifier = 10 kHz.

understanding of field compensation (shimming). Therefore, the shimming procedure must be by trial and error.

Our trial and error procedure of shimming field is as follows.

1. Place one iron screw on an arbitrary position on the shim holder.
2. Check the FID signal. The FID signal with the homemade NMR magnet without averaging is shown in Fig. 5.
3. If the FID signal lasts longer, then accept this screw position. Then, search the second iron screw position. The second position is often near the first one.
4. If the FID signal lasts shorter, then deny this screw position. The "good" position that makes a FID longer is often an axisymmetric position of the "bad" position.

The above protocol was repeated until the longest FID signal was obtained.

It took approximately 2 days for the first shimming. Eleven iron screws were finally used. Once the shimming was done, the homogeneity of the magnetic field did not deteriorate as long as it was kept quiet. In addition, even if the homogeneity deteriorated temporarily, it was possible to recover the same homogeneity within a few hours of shimming by trial and error. As shown in Fig. 6, the FWHM of the peak of the pure water sample was 45 ppm before shimming, but it was improved to 1 ppm after shimming.

3.2. Shimming other NMR magnets

In addition to the above NMR magnet, we applied our iron screw shimming to two other NMR magnets: one is another homemade NMR magnet and the other is a commercial one shown in Fig. 7. We confirmed that our shimming mechanism is equally applicable to any NMR magnets.

The commercial magnet is a part of an NMR system [32]. The equipment shown in Fig. 7 consists of the NMR magnet, the FPGA based signal processing unit, a preamplifier, an rf power amplifier, and a matching circuit. This is controlled via USB. It is an all-in-one NMR system. This commercial NMR magnet was designed according to our reported work [22] and thus it has enough frequency resolution so that the chemical shift of fluorine can be resolved: it is sufficient as an educational device. However, if one can achieve frequency resolution capable of observing the chemical shift of hydrogen, it would be more valuable for the educational purposes.

The iron screw shimming for the commercial NMR magnet was similarly performed as in the case of the homemade NMR magnets because the basic structures are the same. Although the readers may think that the homemade NMR magnets and the commercial one are very different, their structures are essentially the same. See, the right panel of Fig. 3. In other words, our design [22] is very robust against different manufacturing procedures. Six screws were used for shimming. Fig. 8 shows the spectrum of pure water. The solid line represents the spectrum measured immediately after shimming, and the dotted line denotes the spectrum after 15 days. The field homogeneity was stable. This stability is important as an educational device: students do not need to make shimming NMR magnets by themselves.

The spectrum shown in Fig. 9 was measured with 100 % acetic acid sample. The signals from hydrogen nuclear spins in a methyl and carboxy group were resolved here. The FWHM's of these peaks were measured as 2.9 ppm by comparing with the peak distance of 9.7 ppm.

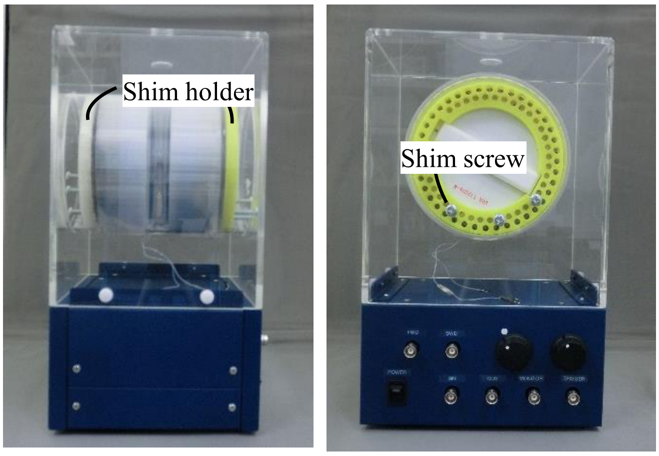

Figure 7: We obtained a commercial NMR magnet without screw shims from THAMWAY Co., Ltd. Shim holders were placed on the commercial NMR magnet.

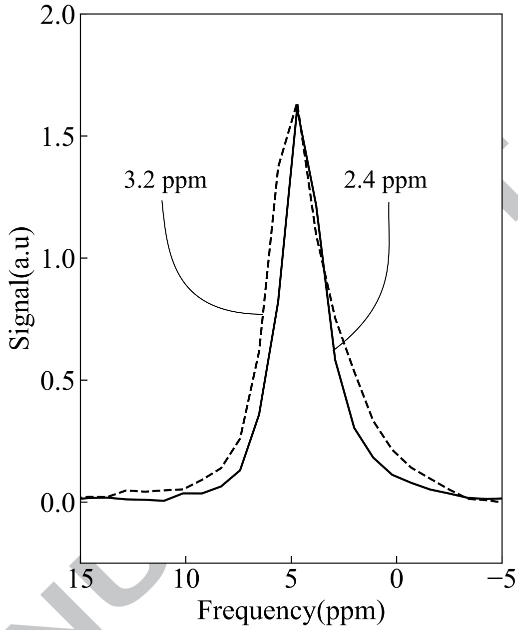

Figure 8: Spectra of pure water with commercial NMR magnet shown in Figure 7. Dotted line represents data measured 15 days after the day when data pertaining solid line was obtained. Homogeneity of magnetic field is stable. Detailed FID measurement conditions are as follows. Solid line: the rf pulse frequency = 2.4648 MHz (57.889 mT), and the cutoff frequency = 1 kHz. The dashed line: 2.4630 MHz (57.847 mT), 1 kHz.

4. Conclusion

The field homogeneity of a simple and low-cost NMR magnet [22] was improved by adding a simple shimming mechanism, or by attaching several iron screws to the back of the ferrite magnets. The homogeneity was improved from about 50 ppm to 1 ppm under the most ideal conditions. With this homogeneity, we successfully resolved two NMR peaks corresponding to hydrogen nuclear spins in a carboxy and methyl group in the acetic acid sample.

Our method reported in this paper may be applicable to magnet systems employed in other magnet systems, such as particle accelerators for elementary particle experiments. Since the magnetic pieces for shimming are arranged outside the magnet, their effects are not large and thus fine tuning is easy with our method.

We are now investigating the detailed mechanism of iron screw shimming by simulating its effects on a magnetic field [31] in order to find a rigorous algorithm for shimming. We are also planning to implement quantum algorithms in our NMR system.

Acknowledgment

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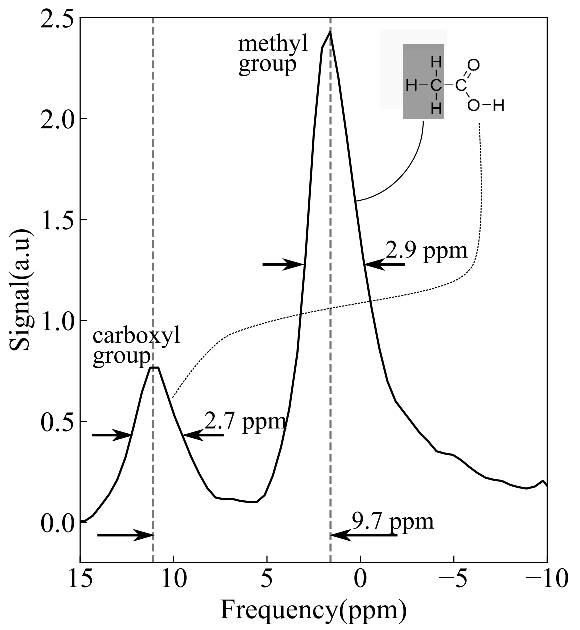

Figure 9: Spectrum of acetic acid measured with commercial NMR magnet after shimming. FWHM's of these peaks are 2.9 ppm for both hydrogen nuclear spins in carboxy and methyl group in acetic acid sample. Measurement condition is as follows. The rf pulse frequency = 2.4644 MHz (57.880 mT). The cutoff frequency = 1 kHz.

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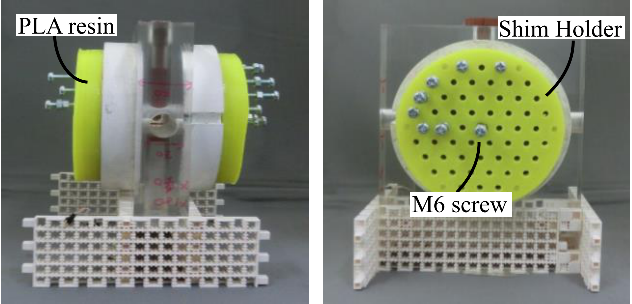
Research highlights

- We improved the resolution of tabletop NMR by shimming with screws.
- Its spectral resolution is 2 Hz at the Larmor frequency of 2.5 MHz.
- Chemical resolution of acetic acid can be confirmed by improving resolution.
- This resolution is obtained by employing shim screw and two pure iron discs with ferrite magnets .
- This is good for building a NMR spectrometer for education.
- The components can be printed by a 3D printer.


PLA resin
Shim Holder
M6 screw